

Michael S

Data Scientist

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Summary

Computer Science student and Machine Learning Engineer specializing in Artificial Intelligence, Computer Vision, and Data Engineering. Engineered and deployed end-to-end ML models, real-time computer vision systems, and cloud-integrated REST APIs using Python, PyTorch, TensorFlow, and modern MLOps practices. Proven track record in problem-solving, feature engineering, and data pipeline optimization.

Education

Modern Academy, Maadi Graduate
Cairo, Egypt

Experience

- Machine Learning Engineer Trainee (Microsoft Track), DEPI – Digital Egypt Pioneers Initiative** 07/2026 - Present
- Training on the Microsoft Machine Learning Engineer track of Egypt's national Digital Egypt Pioneers Initiative, an applied programme focused on building production-oriented machine learning skills
- AI Training Program Participant, Sharks** 07/2023 - 09/2023
- Explored foundational machine learning concepts and supervised classification models, implementing algorithms such as Support Vector Machines (SVM).
 - Formulated data exploration and visualization pipelines, completing a capstone classification project on the Iris benchmark dataset to predict species characteristics.
- AI Training Program Participant, IT Gate** 07/2022 - 09/2022
- Completed an intensive 2-Months foundational training on core Python development and interactive environment setups using Jupyter Notebooks.
 - Practiced foundational data analysis techniques using NumPy and Pandas for data manipulation, structured cleaning, and introductory statistical visualization.

Projects

Football Computer Vision System

Technologies: Python, OpenCV, YOLOv8, ByteTrack, PyTorch

- **Trained** a YOLOv8s object detection model across 3,052 images and 4,835 instances, achieving 84.8% mAP50 and 84.9% Precision.
- **Implemented** ByteTrack multi-object tracking to process real-time match analytics at 25 FPS.

Football Scout REST API

Technologies: Python, Flask, XGBoost, Scikit-Learn, Docker

- **Built** an ML-driven scouting platform analyzing 1,469 professional players, achieving 99.2% recommendation accuracy.
- **Deployed** an XGBoost valuation model ($R^2 = 0.753$, $MAE = \text{€}6.29M$) via a Flask REST API interface.

Expected Goals (xG) Shot Quality

Pipeline+++++

Technologies: Python, Scikit-Learn, Pandas

Designed and evaluated classification models estimating football shot quality and probability

NYC Airbnb Market Data Analysis

Technologies: Python, Pandas, NumPy, Power BI, Excel

- **Cleaned and structured** 40,000+ raw market listings, engineering custom location and pricing feature sets to analyze urban rental trends.
- **Built interactive Power BI dashboards** with slicers, KPI cards, and custom DAX measures to surface spatial pricing patterns and dynamic revenue drivers.

Skills

Programming: Python, C++, SQL, HTML/CSS
Machine Learning and AI: Scikit-Learn, PyTorch, TensorFlow, XGBoost, Random Forest, Feature Engineering
Computer Vision & Data Engineering: OpenCV, YOLO (YOLOv8), Object Detection, Multi-Object Tracking (ByteTrack)

Data and Analytics: Pandas, NumPy, Matplotlib, Seaborn, Power BI (Power Query, DAX)

Databases and Backend: MySQL, REST APIs, Flask

Cloud and Tools: GitHub, VS Code

Soft Skills: Technical Problem-Solving, Analytical Thinking, Data-Driven Strategy, Agile Collaboration, Adaptability

Certifications

- Supervised Machine Learning: Regression and Classification – Stanford / DeepLearning.AI [[Credential](#)]
- Python for Data Science & AI – IBM [[Credential](#)]
- Databases and SQL for Data Science with Python – IBM [[Credential](#)]
- Power BI Business Analytics – UC Davis

Languages

Arabic native ·

English